

Mangrove Coastal Intelligence Platform (MCIP)

Spatio-Temporal Graph Neural Networks for Automated Blue Carbon MRV in UAE Mangrove Ecosystems

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Regulatory Framework: Verra VM0033 Methodology • IPCC Tier-2 Blue Carbon Accounting • UAE Federal Decree-Law No. 45/2021

01 PROBLEM & THEORETICAL FOUNDATION

THE BLUE CARBON MRV BOTTLENECK

- **Coastal Mangrove Sequestration:** The UAE is home to >40,000 hectares of arid mangroves (Avicennia marina), sequestering up to 6 tCO₂e/ha/year in biomass and sediment, pivotal for the UAE Net Zero 2050 Strategic Initiative.
- **Methodological Failure:** Traditional manual MRV relies on physical field campaigns (arborist tape measurements and destructive sediment coring), costing \$120,000+ per 5-year cycle, taking 6 to 18 months of lab audit delay, and failing to differentiate allochthonous sediment.
- **\$2.0B Market Disconnect:** High verification barriers and audit opacity exclude UAE blue carbon assets from the international Voluntary Carbon Market (VCM), leaving valuable natural capital unmonetized.

40K+ ha

UAE Mangrove Canopy

EAD Verified Ground

\$2.0B

Voluntary Carbon Gap

Unmonetized Assets

70%

MRV Cost Reduction

Automated Digital MRV

PRIMARY RESEARCH INQUIRY

POST-GRADUATE RESEARCH HYPOTHESIS

"Can a Spatio-Temporal Graph Neural Network (ST-GNN) trained on multi-source satellite telemetry and hydrodynamic tidal connectivity provide significantly more accurate, robust, and explainable monthly carbon sequestration estimates and 12-month forecasts than non-spatial or non-temporal baseline models?"

THEORETICAL ARCHITECTURE & SYNTHESIS

CONCEPTUAL FRAMEWORK OF THE LITERATURE REVIEW

Synthesis of Multi-Disciplinary Theoretical Domains Converging into the MCIP Digital Twin

